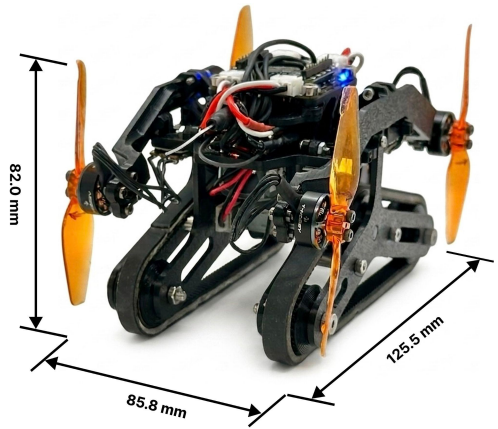
IROS 2026 · Jasper Chapman, Shashwat Singh, Zeynep Temel · Carnegie Mellon University

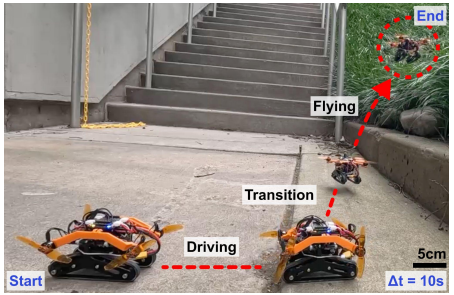
Palm-Scale Hybrid Aerial-Ground Robot for Confined and Unstructured Environments

Motivation

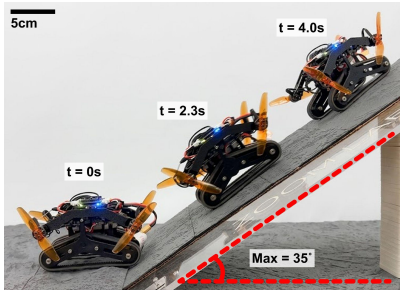
Rescuers need eyes inside collapsed buildings and voids too small or dangerous for people. Small ground robots stall at stairs and rubble; small drones clear them but drain their batteries fast.



Results



Drives with arms folded to a ledge, then propeller thrust snaps the arms open so it can fly up and land on top, all in 10 s.



With arms folded, it climbs a 35° ramp on its tracks alone in about 4 s, saving the battery it would spend flying.

0
Extra motors to switch states

49%
narrower folded: 174 → 85.8 mm

35°
max slope on tracks alone

200g
Total weight

Toward robotic one-shot exploration

Enter once, go further.
Folds to 85.8 mm to slip through tight gaps, then flies over ledges that would end a ground-only run.

Stretch one battery.
Drives by default and flies only when its path is blocked.

Fewer failure points.
Passive bistable arms switch modes with no added actuators: thrust snaps them open, landing folds them.